## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Statistical association, not proof of causation.

Q2. An estimate guiding an algorithm's next choice.

Q3. A quantity to minimise/maximise.

Q4. Five: 32→16→8→4→2→1.

### You Do pages · Independent questions

Q1. B route10 [2], C route6 [2], choose C [1], local first-edge choice misleading [1].

Q2. 1200/80=15s [4]; stable throughput/no overhead or contention [2].

Q3. Fast estimate [1], does not model directed road restrictions [1] or congestion [1], must respect admissibility if A* optimality required [1], validate against real routes [2].

### Exit Ticket · Model answer

False. Future edge costs can reverse the choice.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**



RHS COMPUTING / OCR H446 / mixed LESSON 070 · DRAFT

// BLACK PEN Answer all questions first. //✓ GREEN PEN Then: tick and rewrite mistakes.

# LESSON 070 · Lesson 070 · Heuristics and Performance Modelling

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is correlation?

Q2. What does a heuristic provide?

Q3. What is an optimisation objective?

Q4. A sorted 32-element candidate set is halved to one candidate. How many halvings are needed?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

heuristic — Estimate guiding solution search.

greedy — Choose apparently best immediate option.

performance model — Simplified resource/time estimate.

admissible — Heuristic never overestimates remaining optimal cost.

### New learning

Define and predict — A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

Explain — A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

Evaluate — Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

### Worked example

Choose cheaper complete path: X→Y→G is 3+9; X→Z→G is 6+2.

1. Follow the reasoning — Calculate the entire feasible route rather than the next immediate edge.

## We Do · Practise with guidance

Estimate processing 600 records at 50 records/s, then discuss what model ignores.

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600/50=12s under constant-rate assumptions; ignores setup/IO/congestion.
```

Follow the reasoning

A performance model is a simplified prediction, not a guaranteed elapsed time.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Compare route A-B-G edges 2+8 against A-C-G 5+1. Calculate totals and select cheaper. [6]

### Question 2

Model 1200 jobs processed at 80/s, under constant throughput: estimate duration and one assumption. [6]

### Question 3

Evaluate using straight-line distance as heuristic for car routes with one-way roads and congestion. [6]

## Exit Ticket · Answer before checking

A local greedy choice always produces an optimal complete path.